SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks.* *(BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I VAIBHAVA LAKSHMI K (192511310) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given:

- File size = **150 MB = 150,000,000 bytes**
- Segment size = **1500 bytes**
- Header + Trailer = **40 bytes/frame**
- Bandwidth = **50 Mbps = 50,000,000 bits/s**

(i) Number of Segments

Number of Segments = File Size / Segment Size

= 150,000,000 / 1500

= **100,000 segments**

(ii) Number of Frames

Each segment is encapsulated into one frame.

Therefore,

Number of Frames = **100,000 frames**

(iii) Total Data Link Layer Overhead

Total Overhead = Number of Frames × Overhead per Frame

= 100,000 × 40

= **4,000,000 bytes**

= **4 MB**

(iv) Total Transmission Time

Total Data Transmitted = File Size + Total Overhead

= 150,000,000 + 4,000,000

= **154,000,000 bytes**

Convert bytes into bits:

= 154,000,000 × 8

= **1,232,000,000 bits**

Transmission Time = Total Bits / Bandwidth

= 1,232,000,000 / 50,000,000

= **24.64 seconds**

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Sliding window size = 6 frames
- Total frames = 48
- One frame lost in every window

(i) Number of Transmission Windows

Number of Windows = Total Frames / Window Size

= 48 / 6

= **8 windows**

(ii) Number of Retransmissions

One frame is lost in each window.

Retransmissions = 8 × 1

= **8 frames**

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given:

- IP packet size = 12,000 bytes
- MTU = 1500 bytes
- IP header = 20 bytes

(i) Payload per Fragment

Payload = MTU − Header

= 1500 − 20

= **1480 bytes**

(ii) Number of Fragments

Number of Fragments = 12,000 / 1480

= 8.11

Since a fraction of a fragment is not possible,

Number of Fragments = **9**

(iii) Total Header Overhead

Header Overhead = Number of Fragments × Header Size

= 9 × 20

= **180 bytes**

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Sliding window size = 6 frames
- Total frames = 48
- One frame lost per window

(i) Number of Windows

Number of Windows = 48 / 6

= **8 windows**

(ii) Retransmissions

Retransmissions = 8 × 1

= **8 frames**

5.A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given:

- File size = 600 MB
- Checkpoint interval = 60 MB
- Failure after = 390 MB

(i) Number of Checkpoints

Number of Checkpoints = 390 / 60

= 6.5

Only completed checkpoints are counted.

= **6 checkpoints**

(ii) Data to be Retransmitted

Last completed checkpoint = 6 × 60

= 360 MB

Retransmitted Data = 390 − 360

= **30 MB**

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given:

- Original file size = 900 MB
- Compression = 40%
- Encryption overhead = 5%

(i) Compressed File Size

Compressed Size = 900 × 60%

= **540 MB**

(ii) Encrypted File Size

Encrypted Size = 540 × 1.05

= **567 MB**

(iii) Percentage Reduction

Reduction = ((900 − 567) / 900) × 100

= (333 / 900) × 100

= **37%**

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given:

- Data size = 3 GB = 3000 MB
- Request size = 3 MB
- Throughput = 120 Mbps

(i) Number of Requests

Number of Requests = 3000 / 3

= **1000 requests**

(ii) Transmission Time

Total Data = 3 × 8 × 1000

= 24,000 Mb

Transmission Time = 24,000 / 120

= **200 seconds**

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Given:

- Application Layer = 4 ms
- Transport Layer = 3 ms
- Network Layer = 5 ms
- Data Link Layer = 2 ms
- Physical Layer = 1 ms
- Propagation delay = 18 ms
- Transmission delay = 12 ms

(i) Total Processing Delay

Processing Delay = 4 + 3 + 5 + 2 + 1

= **15 ms**

(ii) Total End-to-End Delay

Total Delay = 15 + 18 + 12

= **45 ms**

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given:

- Useful data = 80 MB = 80,000,000 bytes
- Frame size = 1600 bytes
- Overhead = 80 bytes

(i) Payload per Frame

Payload = 1600 − 80

= **1520 bytes**

(ii) Number of Frames

Number of Frames = 80,000,000 / 1520

≈ **52,632 frames**

(iii) Total Overhead

Overhead = 52,632 × 80

= **4,210,560 bytes**

(iv) Transmission Efficiency

Efficiency = (80,000,000 / (80,000,000 + 4,210,560)) × 100

= **95%**

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given:

- Original file size = 240 MB
- Compression = 30%
- Segment size = 1400 bytes
- Data Link overhead = 60 bytes/frame

(i) Compressed File Size

Compressed Size = 240 × 70%

= **168 MB**

Convert into bytes:

168 MB = **168,000,000 bytes**

(ii) Number of Segments

Segments = 168,000,000 / 1400

= **120,000 segments**

(iii) Total Protocol Overhead

Overhead = 120,000 × 60

= **7,200,000 bytes**

= **7.2 MB**

(iv) Final Data Transmitted

Final Data = 168 + 7.2

= **175.2 MB**

RUBRICS FOR CALCULATION

Criteria	Weighta g e	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginn ing
Understandin g of Concepts	25%	Demonstrates a thorough, accurate understandin g of concepts	Good understanding with minor gaps	Basic understanding with some misconception s	Poor understandin g; major misconceptio n s
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organizati on, incoherent answers
		and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandabl e with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand